



Two-step hydrothermally synthesized Ce_{1-x}Zr_xO₂ for oxidative dehydrogenation of ethylbenzene with carbon dioxide

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ABSTRACT

Ceria, zirconia, Ce_{0.9}Zr_{0.1}O₂, Ce_{0.75}Zr_{0.25}O₂, and Ce_{0.5}Zr_{0.5}O₂ samples were synthesized via the two-step hydrothermal method by using Ce(NO₃)₃·6H₂O and Zr(NO₃)₄·5H₂O as the precursors, respectively. The oxides were applied as catalysts for the dehydrogenation of ethylbenzene by using carbon dioxide as a soft oxidant (CO₂-ODEB) at 550 °C and 0.1 MPa. Reaction results indicate that ceria was the most active catalyst while the zirconia with a cubic structure was inactive for CO₂-ODEB. In the cases of the oxides by inserting Zr into CeO₂, the activity indexed by either the initial ethylbenzene (EB) conversion or the calculated turnover frequency (TOF) at a time on stream of 1 h was changed in the sequence of Ce_{0.9}Zr_{0.1}O₂ > Ce_{0.75}Zr_{0.25}O₂ > Ce_{0.5}Zr_{0.5}O₂ while the decreasing pattern of the stability was observed in the order of Ce_{0.5}Zr_{0.5}O₂ ≈ Ce_{0.75}Zr_{0.25}O₂ > Ce_{0.9}Zr_{0.1}O₂. As revealed from the characterization results of XRD and Raman, the pure cubic solid solution of Ce_{1-x}Zr_xO₂ without any phase separations was obtained irrespective of the compositions. By correlating the characterization results, the activity of Ce_{1-x}Zr_xO₂ for CO₂-ODEB was determined by the amount of the Ce³⁺ oxidized by CO₂ and the rate of Ce³⁺/Ce⁴⁺ redox cycles. Moreover, both the coking rate and the promoting effect of the deposited coke on CO₂-ODEB were responsible for the stability of Ce_{1-x}Zr_xO₂.

1. Introduction

As one of the most important raw materials in the petrochemical industry, styrene (ST) is extensively used for the production of rubbers and plastics [1]. Commercially, styrene is mainly produced via the direct process, in which ethylbenzene (EB) is dehydrogenated in the atmosphere of the over-heated steam. Due to the thermodynamic limitation and the large latent heat of water, the process characterizes in a high energy cost [2–4]. To overcome the shortcoming of the direct EB dehydrogenation, alternative methods of oxidative EB dehydrogenation are proposed by using different oxidants such as O₂, N₂O, or CO₂ [5,6]. Among these alternative routes, the oxidative dehydrogenation of EB with CO₂ (CO₂-ODEB) has attracted much attention as a result of the chemical utilization of the greenhouse gas, the mild oxidation ability of CO₂, and the thermodynamically favoured dehydrogenation [2,7–9]. Although much effort has been paid for CO₂-ODEB [9–18], the lack of a stable catalyst with a reasonable activity is still a challenge for the commercialization of CO₂-ODEB [14,19].

Although the surface reaction mechanism at an atomic level is still

unclear, two general mechanisms, i.e., the coupling of the direct dehydrogenation and the reverse water-gas shift reaction and the Mars-van Krevelen redox mechanism, are exclusively discussed for CO₂-ODEB [8,20]. Over the typical oxidation catalyst such as vanadia or ceria, it is commonly accepted that CO₂-ODEB very probably proceeds via the reduction/oxidation (redox) mechanism [7,21–23]. According to the Mars-van Krevelen redox mechanism, the activated EB molecules react with the lattice oxygen over the metal oxide leading to the products of ST and H₂O. Simultaneously, the consumed lattice oxygen (oxygen vacancy) in the oxide is replenished by reacting with CO₂ accompanied with the release of CO. Thus, the rate and the balance of the reduction and oxidation steps participated by the lattice oxygen as an intermediate is expected to play a crucial function in deciding the catalytic activity and stability of CO₂-ODEB. In this case, if the amount and mobility of the removable lattice oxygen or oxygen vacancy and its reactivity with CO₂ are high, an active and stable catalyst for CO₂-ODEB is reasonably expected.

The reasonable world reserves, rich structure with different doping cations, and high oxygen storage capacity (OSC) make ceria an

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